Jui-Te 'Ray' Huang

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Education

Carnegie Mellon University **Pittsburgh, PA**
PhD in Robotics **Current**
MS in Robotics (MSR) **May 2024**

National Yang Ming Chiao Tung University **Hsinchu, Taiwan**
BS, MS in Electrical and Computer Engineering **Aug. 2021**

Experience

Research Assistant, Carnegie Mellon University **Sep. 2022 – present, Pittsburgh, PA**
Robot Perception Lab, advised by Prof. Michael Kaess.

- Radar-Inertial State Estimation and Mapping using mmWave Imaging Radar.
  - Research on robust state estimation in adverse environmental conditions.
  - Research on learning-based radar signal processing for state estimation.
  - Implemented radar-visual-inertial sensor fusion state estimation.
  - Developed a large-scale outdoor SLAM system using multiple high-resolution imaging radars.
- GPU-accelerated radar signal processing.
  - Implemented a MIMO signal processing pipeline that generates radar point clouds from raw digital signals.
  - Implemented Synthetic Aperture Radar algorithms to create mmWave radar occupancy maps.
- Built a multi-modal sensor rig (radar, LiDAR, stereo RGB/thermal/event cameras) and co-led a large-scale data collection campaign.

Research Intern, Bosch Research USA **May 2026 – Aug. 2026, Pittsburgh, PA**
May 2025 – Aug. 2025, Pittsburgh, PA

- Developed a conditional latent diffusion model that synthesizes complex-valued, multi-antenna range–Doppler radar spectra from lidar, camera, ego-motion, and radar waveform parameters.
- Used a phase-preserving spectrum autoencoder with a flow-matching diffusion transformer, so generated spectra can be processed with standard radar signal processing and pretrained networks.
- Generated raw radar spectra from CARLA simulator scenes; augmenting radar foundation model training with generated data gave gains on the I/Q-1M dataset equivalent to $2\times$ data scaling.
- Built a large-scale training and evaluation pipeline; paper in submission.

Research Intern, University of Washington **Sep. 2021 – Feb. 2022, Seattle, WA**
Information Processing Lab, advised by Prof. Jenq-Neng Hwang.

- Built and calibrated a sensor system to collect synchronized data for radar multi-object tracking.
- Learning-based multi-object detection and tracking using mmWave imaging radar.
- 3D multi-object tracking using LiDAR and visual depth estimation.

Research Assistant, National Chiao Tung University **Jun. 2018 – Aug. 2021, Hsinchu, Taiwan**
Assistive Robotics Group, advised by Prof. Hsueh-Cheng Wang.

- DARPA Subterranean Challenge, SLAM and Deep Reinforcement Learning Navigation.
  - Research on navigation using sim-to-real deep reinforcement learning.
  - Research on cross-modal learning to navigate robots through smoke using mmWave radar.
  - Team NCTU, Software Team Lead and Human Supervisor in SubT Challenge: Urban Circuit.
  - Implemented a LiDAR SLAM system for large-scale underground environments.
- Maritime RobotX Challenge, Autonomous Surface Vehicle.
  - Built the sensor and computation system for an autonomous boat (WAM-V).

Research Interests

Radar Perception, SLAM, State Estimation, Generative Sensor Simulation, Multi-Modal Learning, Field Robotics

Selected Projects & Publications

- **Radar-Inertial Odometry and Mapping**
  - **Jui-Te Huang**, Tianshu Huang, Anthony Rowe, and Michael Kaess, "UNRIO: Uncertainty-Aware Velocity Learning for Radar-Inertial Odometry," in submission, 2026. **Best Student Poster**, ICRA 2026 Radar in Robotics Workshop.
  - **Jui-Te Huang***, Ruoyang Xu*. Örebro 4D-Radar SLAM Challenge, **1st place**. ICRA 2026 Radar in Robotics Workshop.
  - **Jui-Te Huang**, Ruoyang Xu, and Michael Kaess, "Millimeter Wave Radar: From Synthetic Aperture to Probabilistic Mapping," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
  - **Jui-Te Huang**, Ruoyang Xu, Akshay Hinduja, and Michael Kaess, "Multi-Radar Inertial Odometry for 3D State Estimation using mmWave Imaging Radar," *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
- **Radar Multi-Object Tracking for Autonomous Driving**
  - Y. Wang, J.-H. Cheng, **J.-T. Huang**, S.-Y. Kuan, Q. Fu, C. Ni, S. Hao, G. Wang, G. Xing, H. Liu, and J.-N. Hwang, "Vision meets mmWave Radar: 3D Object Perception Benchmark for Autonomous Driving," *IEEE Intelligent Vehicles Symposium (IV)*, 2023.
- **Search and Rescue in DARPA Subterranean Challenge**
  - C.-L. Lu*, **J.-T. Huang***, et al., "A Heterogeneous Unmanned Ground Vehicle and Blimp Robot Team for Search and Rescue using Data-driven Autonomy and Communication-aware Navigation," *Field Robotics, Special Issue: Advancements and Lessons Learned during Phase I & II of the DARPA Subterranean Challenge*, 2022. (*Equal contribution)
- **Cross-modal Learning and Reinforcement Learning for Navigation**
  - **J.-T. Huang**, C.-L. Lu, P.-K. Chang, C.-I Huang, C.-C. Hsu, Z. L. Ewe, P.-J. Huang, and H.-C. Wang, "Cross-Modal Contrastive Learning of Representations for Navigation using Lightweight, Low-Cost Millimeter Wave Radar for Adverse Environmental Conditions," *IEEE Robotics and Automation Letters (RA-L)*, 6(2), 3333–3340, 2021.
  - C.-L. Lu, Z.-Y. Liu, **J.-T. Huang**, C.-I Huang, B.-H. Wang, Y. Chen, N.-H. Wu, H.-C. Wang, L. Giarré, and P.-Y. Kuo, "Assistive Navigation using Deep Reinforcement Learning Guiding Robot with UWB/Voice Beacons and Semantic Feedbacks for Blind and Visually Impaired People," *Frontiers in Robotics and AI*, 2021.

Academic Service

- Teaching Assistant, CMU: 16-833 Robot Localization and Mapping, 16-822 Geometry-based Methods in Vision.
- Teaching Assistant, NCTU: Sensing and Intelligent Systems, Human-Centric Computing Lab, Introduction to Artificial Intelligence.
- Reviewer for IEEE IROS, ICRA, RA-L, T-ASE.

Skills

- **Programming Languages:** C++, Python, CUDA
- **Software, Middleware, and Libraries:** ROS, Docker, OpenCV, Open3D, PCL
- **Deep Learning / Optimization Frameworks:** PyTorch, JAX, GTSAM
- **Manufacturing:** Autodesk Fusion, KiCAD, 3D Printing, PCB Design